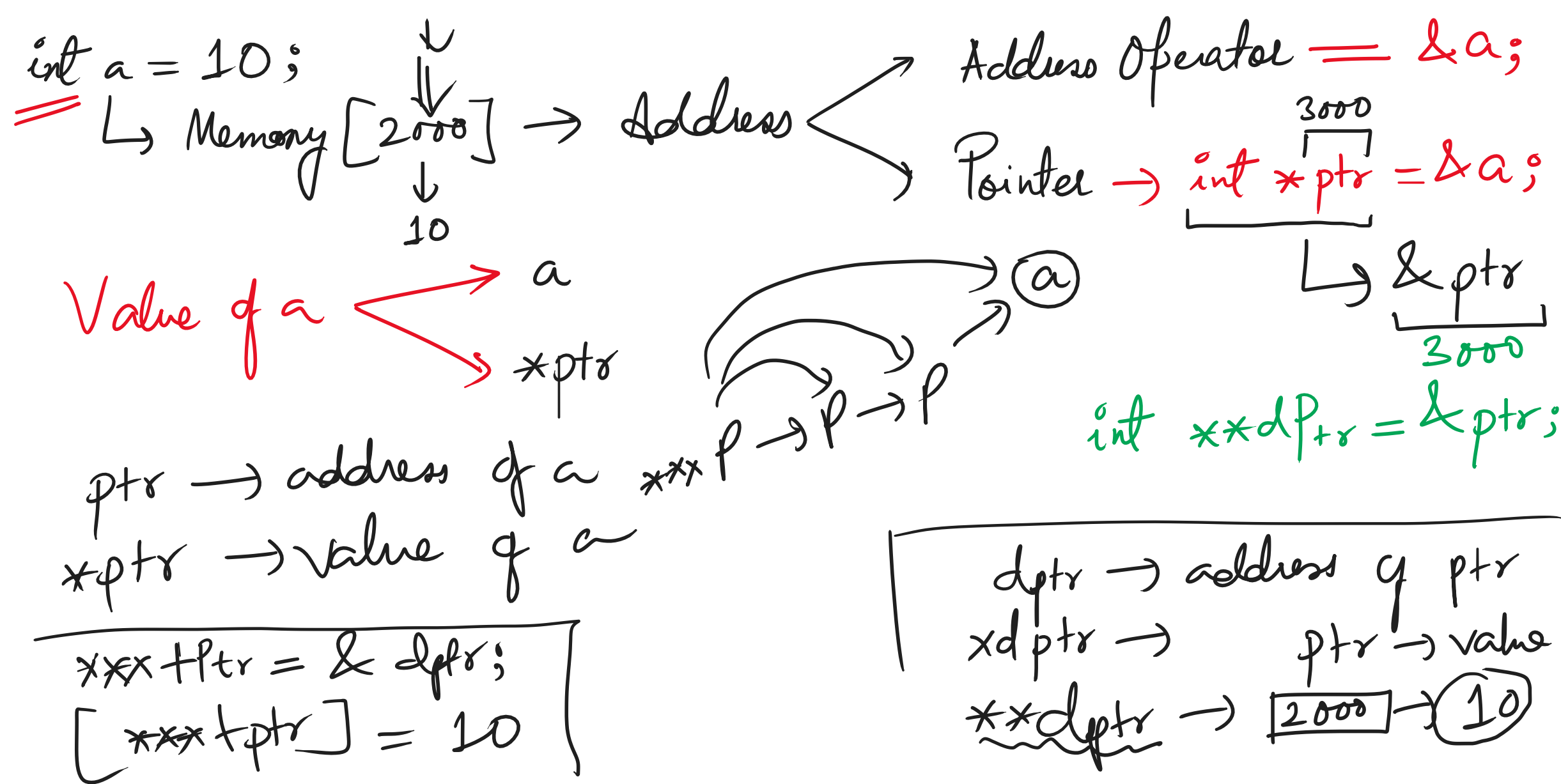




C/C++/Java: Array → Homogeneous Collection of Data

Python data = [1, 2.5, True, 'java'] Heterogeneous

int arr[] = {2, 9, 6, 1, 0, 5};

arr[elements] → arr[3]

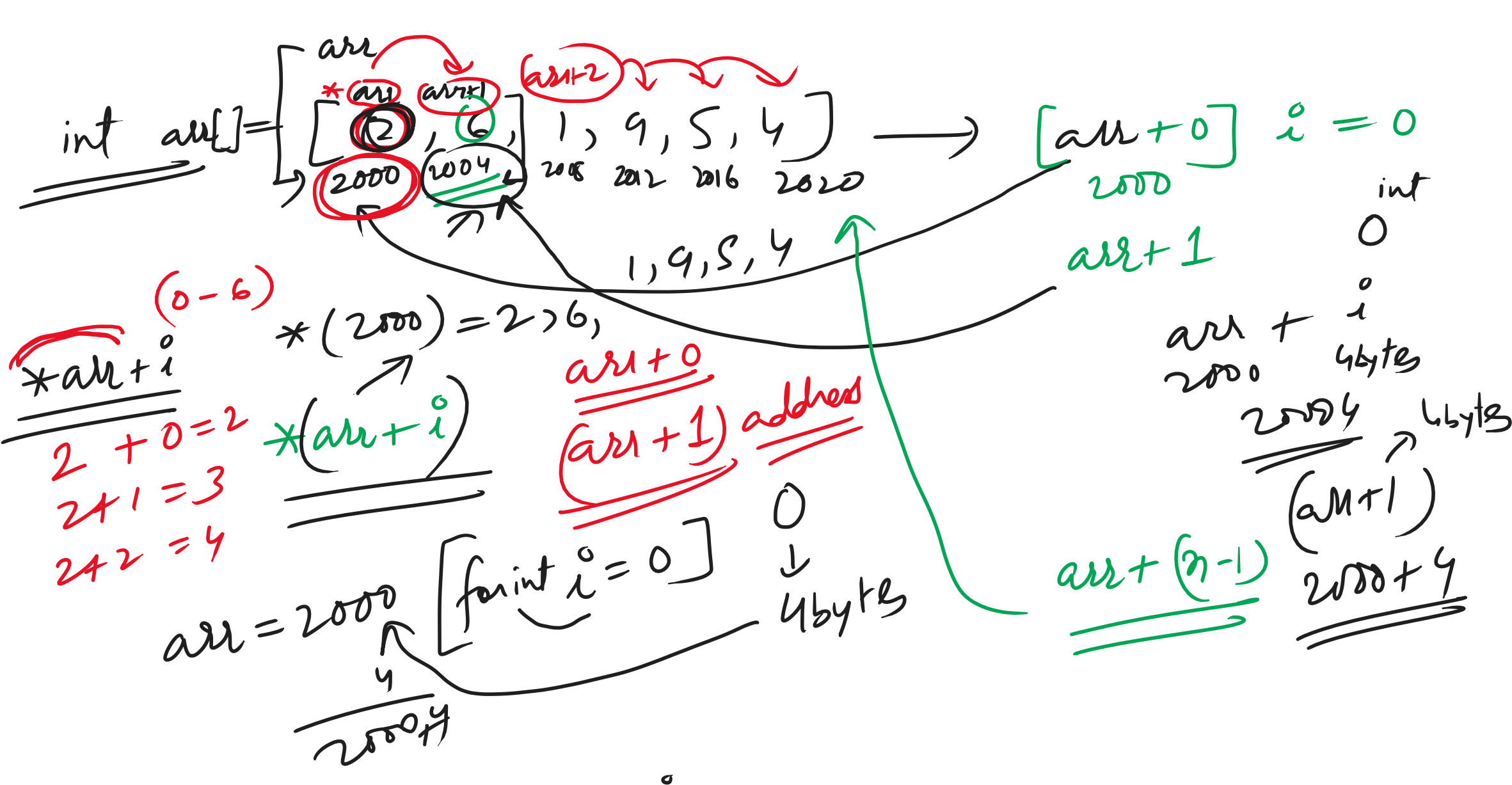
Size = sizeof(arr) / sizeof(arr[0])

int arr[] = {2, 4, 6, 8, 0};

* The name of the array points to the address of the 1st element of the array.

arr → 2000

Array Pointer → arr → 1st element pointer.



* Pattern: 2D

Identity Matrix: r == c → 1

H.W. 4PM

Searching & Sorting Algos

* Linear

* Binary

* Recursive BS

* Interpolation

* Jump Search

* Bubble Sort

* Selection Sort

* Insertion Sort

* Heap Sort

* Sparse Matrix:

No. of elements: 4x5

10 rxc = 20

if {ctr} > (rxc/2)

SM

NSM

Transpose of a Matrix: r → 1 2 3 4

c → 5 6 7 8

1 5

2 6

3 7

4 8

16

Upper Triangle

Lower Triangle

* Trans

* row becomes col

* col becomes row

UTM:

1 2 3 4

5 6 7 8

9 10 11 12

13 14 15 16

1 5

2 6

3 7

4 8

16

Upper Triangle

Lower Triangle

* Dynamic Memory Allocation:

→ malloc()

C → calloc()

C++ → realloc()

→ free()

C + C++ = Java;

C++ → new delete

DMA in C++

* Why is DMA needed?

11:00 to 1:00

2:00